Lecture 28

- HW-9 due coming Friday 11 pm
- HW 7–10 will be on Midterm-III
- UT Sage
- Midterm-III make-up time poll
- Midterm-III practice problems

- DGCX47

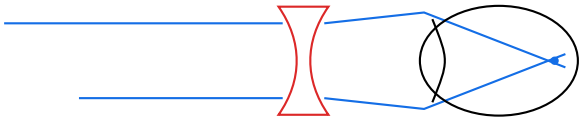
Goals:

1. Lens maker's formula
2. Interference of waves

$$\text{Power} = \frac{1}{f} \qquad \begin{matrix} (f \text{ is in m}) \\ \left(\begin{matrix} \text{Power is in} \\ \text{diopters} \end{matrix} \right) \end{matrix}$$

Q) Power $> -0.25\,\text{D}$

What is the min far point of the eye?



$$d_o = \infty \qquad d_i = -d_{\text{far}}$$

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$\frac{1}{f} = \frac{1}{\infty} - \frac{1}{d_{\text{far}}}$$

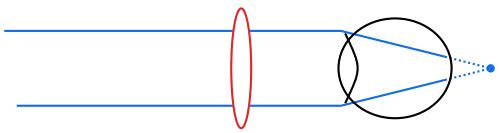
$$f = -d_{\text{far}}$$

$$P = \frac{1}{f} = -\frac{1}{d_{\text{far}}}$$

$$-\frac{1}{d_{\text{far}}} > -0.25$$

$$\frac{1}{d_{\text{far}}} < 0.25$$

$$4\,\text{m} < d_{\text{far}}$$



Lens maker's formula

– General formula for determining focal length of non-thin lenses

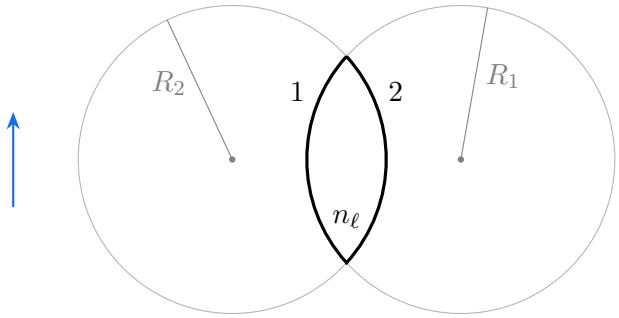
$$\frac{1}{f} = \left(\frac{n_{\text{lens}}}{n_{\text{medium}}} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

IMP — Indices of refraction — Radii of curvature

$$P = \frac{1}{f}$$



Double convex | Plano convex | Meniscus | Meniscus | Plano concave | Double concave



Center of sphere is on left: $R < 0$
" right: $R > 0$

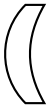
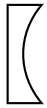
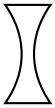
Above example: $R_1 > 0$
$R_2 < 0$

$f > 0$: Converging lens
$f < 0$: Diverging lens

Alternate heuristic

- If side towards object is convex $\Rightarrow R > 0$

$$\begin{pmatrix} \text{front of spoon} = \text{convex} \\ \text{back of spoon} = \text{concave} \end{pmatrix}$$

$R_1 > 0$	$R_1 = \infty$	$R_1 < 0$	$R_1 > 0$	$R_1 = \infty$	$R_1 < 0$
$R_2 < 0$	$R_2 < 0$	$R_2 < 0$	$R_2 > 0$	$R_2 > 0$	$R_2 > 0$
$f > 0$	$f > 0$	$f = ?$	$f = ?$	$f < 0$	$f < 0$
					
Double convex	Plano convex	Meniscus	Meniscus	Plano concave	Double concave

- Lenses thinner in the middle are diverging
- Lenses thicker in the middle are converging

26.8 Lensmaker

The two surfaces of a lens have radii of $r_1 = -20$ *cm* and $r_2 = +5$ *cm*. The index of refraction of the lens is $n = 1.52$. What kind lens is it?

A) Converging

B) Diverging ✓

C) Not enough information

$$\frac{1}{f} = \left(\frac{n_{\text{glass}}}{n_{\text{air}}} - 1\right)\left(-\frac{1}{20} - \frac{1}{5}\right) = 0.52\left(-\frac{5}{20}\right) \qquad \underline{\underline{f < 0}}$$



Interference of waves

– Waves are a disturbance in a medium that transmit information/energy through that medium

$$v = f\,\lambda$$

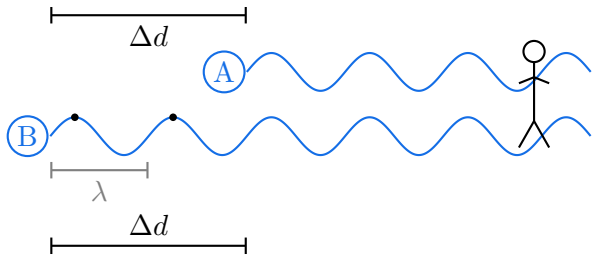
- v: wave speed determined by medium
- f: Frequency
- λ: Wavelength in medium

Constructive: Waves overlap in phase, amplitudes add

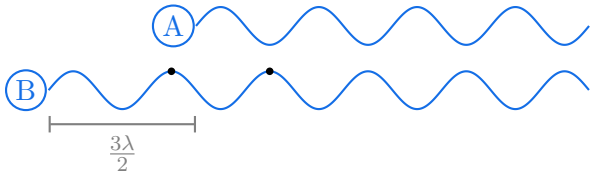
Destructive: Waves overlap out of phase, amplitudes subtract

– Sources (A) & (B) emit waves in phase

Constructive interference



Destructive interference



$$\Delta d = \begin{cases} m\lambda & \text{Cons.} \\ \left(m + \frac{1}{2}\right)\lambda & \text{Dest.} \end{cases}$$

Demo link: `https://phet.colorado.edu/sims/html/wave-on-a-string/latest/wave-on-a-string_all.html`